

A GIS-Based Framework for Enhancing Walkability Through Citizen Participation and AI Insights



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Abstract The rapid growth of cities has increased reliance on automobiles, resulting in several challenges. This research focuses on enhancing walkability in urban environments to address these issues. This study employs a multi-faceted approach, starting with a comprehensive survey to understand current walking habits, safety perceptions, and community design considerations. The core of this research involves utilizing GIS tools, such as ArcGIS and CityEngine, to analyze urban infrastructure, safety, public amenities, and accessibility factors. Patterns derived from this analysis inform the development of recommendations. The research integrates wider sidewalks, marked crosswalks, traffic-calming measures, shaded areas, seating, public transportation integration, bicycle infrastructure, open public spaces, and enhanced safety features into the urban design. The findings suggest that thoughtful urban planning, incorporating AI-driven insights and community feedback, can create environments conducive to walking, thus contributing to healthier, more sustainable cities. The research provides practical recommendations for urban planners, policymakers, and designers to foster pedestrian-friendly urban spaces.

Keywords Walkability · 15-Minute city · Saudi Arabia · Pedestrian mobility · Smart cities · GIS · AI · Urban planning

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1 Introduction

Decisions have been made about urban design, which profoundly impacted the constructed world we currently occupy [1, 2]. In recent years, a growing emphasis has been placed on building safe and environmentally sustainable neighborhoods and improving the walkability of metropolitan areas. According to research conducted by Frank and Sallis [3], a walkable and environmentally conscious design has many advantages. These benefits include improved physical health, a reduction in the amount of traffic that is generated by cars, and more involvement in community activities [1]. According to Gehl (2011), the design of urban environments can significantly impact people's levels of physical activity and social consequences like the rate of criminal activity and the degree to which communities are cohesive [3, 4]. Because of this, urban planners and designers need to prioritize the ability to move around and design in a way that is responsible for the environment.

Walking is widely acknowledged as one of the most effective types of moderate-to-vigorous physical activity [5]. In contrast to more strenuous forms of physical activity (PA), walking is safe for persons of any age [6]. The extent to which the built environment supports and encourages walking by providing safety, high accessibility and connectivity to destinations, and visual interest within a reasonable period is how walkability is described. Walking is the most widespread physical activity and the oldest form of human mobility [7]. It helps to dramatically reduce the number of obese people living in today's highly consuming civilizations [4]. To promote walkability, the "5D" theory, which includes population density, pedestrian-friendly design, diversity of destinations, destination accessibility, and distance to transit, is proposed as a measure of urban Sustainability [6]. The original walkability index is proposed to include three variables: street connectivity, net residential density, and land use mix, which respectively describe the pedestrian-friendly design, population density, and diversity of neighborhood land use [6].

The processing power of computers and other technologies, such as artificial intelligence (AI), continues improving, paving the way for discovering new avenues to solve urban issues [8, 9]. In recent years, AI has attracted substantial attention and utilized in various industries, including healthcare, finance, transportation, and entertainment. The continued development of AI has the potential to bring about a revolution in several business sectors, an increase in productivity, and an improvement in the quality of life for individuals [9–11]. The application of artificial intelligence (AI) has had a transformative impact on many academic disciplines, and urban design is not an exception [10]. Using AI technology in planning and designing urban environments can rationalize decision-making processes, raise efficiency levels, and encourage the growth of sustainable communities.

Urban planning is a complex and ever-evolving field that aims to create sustainable and livable cities [10, 12]. With the advancements in artificial intelligence (AI), various AI applications have emerged to facilitate and optimize urban planning processes. These applications encompass various functionalities such as simulating urban dynamics, predicting pedestrian movements, generating realistic 3D

city models, and analyzing transportation systems [13, 14]. While AI presents the most recent wave in computational urban planning and design approaches, comprehensive scientific treatment and impact evaluation are still needed. The attempts to explore the potential of Machine Learning and Autonomous Reasoning in support of urban planning and management tasks are disparate and fragmented [15]. After 1980, with new computational software and hardware advent, urban design became a more pressing issue. There is a pressing need for urban design simulation technologies that can help explain design city solutions. The term “simulation” describes a depiction of the environment that changes over time [13, 16, 17]. This research compares several popular AI tools frequently utilized in urban planning: Urban Sim, SimWalk, CityEngine, Legion, and Transim [14, 18–20].

Overall, the application of AI technology in urban planning has the significant potential to improve the efficiency of decision-making, enhance the distribution of resources, and encourage the growth of more sustainable communities. This overview focused on applying generative adversarial networks and machine learning methods in urban design. As the field continues to advance, it is crucial for urban designers and planners to successfully harness AI to manage difficult urban challenges and develop livable cities.

2 Methodology

The methodology employed in this study harnesses the power of GIS for a data-driven approach to urban design, ensuring that any AI-driven interventions are grounded in an accurate and comprehensive understanding of the spatial dynamics at play within the selected neighborhood. After ArcGIS mapping and adding variables, analyzing the matrix data, and creating patterns and recommendations, the next step is to use CityEngine to create design recommendations. Write the methodology of using CityEngine and the steps.

Esri CityEngine is a powerful 3D modeling software application that urban planners, architects, and GIS professionals use to create, analyze, and visualize urban environments in 3D based on GIS data. After completing the GIS mapping and analyzing the variable matrix data, CityEngine can be used to create design recommendations.

2.1 Case Study Analysis

This study targets neighborhoods with existing infrastructure conducive to walkability and areas with potential development. The selected neighborhoods for this case study are a district in Yanbu, chosen based on several strategic criteria that align with the research objectives, including (A) The district is notable for its functional

significance, housing essential amenities such as colleges, schools, malls, and hospitals. This concentration of facilities indicates a high potential pedestrian catchment area, making it a relevant study site for walkability enhancement. (B) The area is characterized by its organized urban layout. The structured arrangement of streets, blocks, and public spaces provides a foundational grid integrating AI-driven walkability improvements. (C) The presence of traffic lights and sidewalks suggests that Traffic lights and sidewalks suggest the neighborhood already has basic pedestrian infrastructure. This existing framework offers a tangible starting point for further AI optimization and enhancement. (D) The closeness of amenities contributes to the area's natural walkability. Short distances between key destinations make the neighborhood a prime candidate for demonstrating the effectiveness of AI in incrementally improving the pedestrian experience. (E) Despite its strengths, the neighborhood lacks certain variables critical for comprehensive walkability. This gap presents an opportunity for AI to contribute meaningfully to urban design by identifying and prioritizing areas for development.

2.2 Data Collection and Field Observation

A systematic data collection process is implemented, gathering information on current pedestrian pathways, traffic patterns, and the location of amenities. GIS mapping is used to visualize and analyze the spatial distribution of these elements. The data is then pre-processed to ensure compatibility with AI algorithms, including normalization and resolving any discrepancies. The output from the AI analysis is used to propose specific interventions to enhance walkability. Proposals are assessed for feasibility and impact, considering immediate and long-term effects on the neighborhood's walkability. An iterative approach is adopted, where the AI model is refined through feedback cycles and re-evaluation, ensuring that the solutions remain aligned with the evolving dynamics of the neighborhood.

The methodological framework for the field observation phase involves a structured, time-sensitive approach to data collection and matrix development. Fieldwork was conducted over a one-week period, capturing data across different times to understand temporal variations in walkability. Observations were made during the morning and evening to note pedestrian and vehicular traffic patterns. Nighttime observations indicated very low pedestrian activity, prompting further analysis into the potential causes, such as lack of lighting or perceived safety concerns. Daytime observations revealed pedestrian movement primarily in the morning, with a significant drop from noon until 5 PM, likely due to climatic factors such as high temperatures during midday that deter walking.

2.3 Data Analysis

Each variable within the matrix is assigned a binary value (1 for presence, 0 for absence), with additional notes on condition and usability where relevant. The gathered data is then input into the AI analysis model, which assesses the variables in relation to pedestrian movement patterns and identifies areas for potential enhancement. Qualitative data from field observations, such as pedestrian behavior and environmental conditions, are incorporated into the model. The field observation and variable matrix construction are essential components of the research methodology, laying the groundwork for a data-driven analysis that informs AI-driven interventions to improve walkability. This method ensures that the findings and recommendations are grounded in the actual usage patterns and conditions of the neighborhood, reflecting the genuine needs and behaviors of its residents.

The geographic scope focuses on crucial nodes known as ‘X sites,’ which are thought to be important in influencing pedestrian traffic and access shown in (Fig. 1). These nodes serve as anchors for our study, giving focal points for contextualizing the walkability scores.

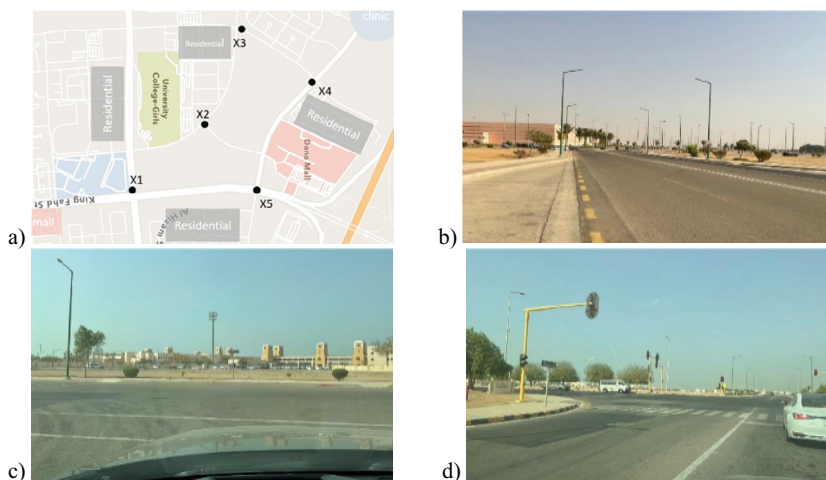


Fig. 1 a Google Map of Yanbu; b Intersection X2, overlooking the Dana mall; c Intersection X3, overlooking the university; d Intersection X5, between Hospital and Dana Mall

3 Results and Discussion

This study focuses on analyzing variable observations made during fieldwork and GIS mapping. We examine the infrastructure, safety measures, amenities, and accessibility factors influencing pedestrian movement. The observations offer a tangible assessment of the physical elements impacting walkability.

3.1 GIS Mapping and Analysis for Walkability Assessment

The pivotal step in assessing the walkability of urban environments involves a detailed spatial analysis using Geographic Information Systems (GIS). For this study, a targeted area within the Saudi Arabian urban context, Yanbu City, has been selected as the subject for an in-depth examination of pedestrian infrastructure and accessibility. By employing ArcGIS, a powerful tool for mapping and analyzing geographic information, this study has integrated various data layers to create a variable matrix that can effectively measure walkability.

This process commenced with carefully delineating the study area boundaries, followed by integrating diverse datasets, including land use patterns, pedestrian pathways, public transit access points, and the distribution of amenities. The geographic scope focuses on key nodes identified as ‘X points’, which are hypothesized to significantly influence pedestrian movement and access, as shown in Fig. 2. These nodes act as anchors for our analysis, providing focal points around which the walkability metrics are contextualized.

The X markers, X1 through X5, appear to be strategically placed at intersections or points of interest, such as near the university (X1), the medical center (X7), and along what seems to be main roads (X2, X3, X4, X5), highlighting critical nodes that would be relevant in assessing walkability. The residential areas marked in yellow surround these amenities, which suggests that the residents have potentially walkable access to key services and institutions. The analysis of the Yanbu district’s current state reveals a well-equipped urban area with essential amenities. However,

Fig. 2 Map of Yanbu, ArcGIS pro program



there are notable opportunities for improvements to walkability, particularly in the distribution of parks and the utilization of central open spaces. Enhancing these aspects could significantly improve the pedestrian experience, making the district functionally comprehensive and more walkable and sustainable.

The variable matrix added to ArcGIS encompasses a range of walkability indicators such as pedestrian infrastructure, safety, urban design, traffic conditions, public transportation, amenities, and accessibility. This matrix is designed to be robust and multifaceted, reflecting the complex nature of urban walkability, which is influenced by a web of interrelated factors. Integrating these indicators into a GIS environment ArcGIS allows for a layered and nuanced analysis, enabling the visualization and quantification of walkability attributes within the urban fabric.

3.2 Variable Matrix Analysis

The variable matrix analysis provides a comprehensive overview of the walkability assessment, capturing key aspects of pedestrian infrastructure, safety, urban design, traffic conditions, public transportation, amenities, and accessibility. This matrix serves as a valuable tool to evaluate and compare the walkability of different locations, represented here as X1 to X5, which correspond to specific areas or neighborhoods. Each cell within the matrix contains manually assessed values, reflecting the condition or presence of various walkability-related variables. Table 1 is a dataset representing various urban variables categorized under pedestrian infrastructure, safety, traffic conditions, public transportation, amenities, and accessibility. Each variable has been assessed for five key points, X1 through X5, within the study area and given a binary value where 1 indicates the presence of the feature and 0 indicates its absence.

Sidewalks are universally present at all nodes, indicating a foundational consideration for pedestrians in urban planning. However, the complete absence of bike lanes signals a notable gap, potentially deterring cycling as an eco-friendly mode of transportation and suggesting an area ripe for development. Marked crosswalks and smart signal crossings, limited to nodes X4 and X5, highlight disparities in pedestrian safety measures that warrant further attention in other parts of the district. The presence of bus stops at nodes X1 and X4 suggests these areas are recognized as transit corridors, integrating pedestrian pathways with public transport options. Conversely, providing car rest stops at X1, X2, and X3 may reflect a vehicular focus in these regions.

The absence of reported accidents is a positive sign. Yet, it does not paint a full picture of pedestrian safety, necessitating a deeper dive into traffic dynamics and enforcement of speed limits, which are commendably in place across the district. The lack of data on public transportation frequency and availability could point to a disconnect in urban design, potentially undermining the viability of walking as part of a multimodal transport network.

Table 1 Variable matrix showing walkability assessment scores (X1 to X5)

Variable category	Variable description	X1	X2	X3	X4	X5
Pedestrian infrastructure	Sidewalks	1	1	1	1	1
	Bike lane	0	0	0	0	0
	Marked crosswalk	0	0	0	1	1
	Signage cross	0	0	0	0	0
	Smart signal cross	0	0	0	1	1
	Signage stops	0	0	0	0	0
	Stations bus	0	0	0	0	0
	Bus stops	1	0	0	1	0
	Car rest stop	1	1	1	0	0
Safety	Car accident	0	0	0	0	0
	Pedestrian accident	0	0	0	0	0
	Traffic light	0	0	0	1	1
	Smart traffic light	0	0	0	1	1
Traffic conditions	Traffic congestion	0	0	0	0	0
	Speed limits	1	1	1	1	1
	Pedestrian-signals	0	0	0	0	0
Public transportation	Availability of transit	0	0	0	0	0
	Transit frequency	0	0	0	0	0
Amenities	Parks and green spaces	1	1	1	1	0
	Restrooms	0	0	0	0	0
	Public seating	0	1	0	0	1
	Shops and services	0	0	0	1	1
Accessibility	Access to amenities	1	1	1	1	1
	Pedestrian accessibility	1	1	1	1	1

Nodes with parks and green spaces, except for X5, suggest areas for leisure and respite that enhance the walking experience. However, limited public seating indicates a missed opportunity for rest and social interaction, which could particularly affect vulnerable populations. The concentration of shops and services at X4 and X5 aligns with these nodes as potential commercial centers, likely encouraging foot traffic and local commerce. Overall accessibility is reportedly good, yet further qualitative analysis is necessary to validate pedestrian routes’ practicality and connectivity to amenities. This pattern analysis underscores the need for an integrated approach to urban design that prioritizes comprehensive pedestrian infrastructure, including cycling paths and well-distributed amenities, to cultivate a truly walkable community.

4 Conclusion

This research has conclusively demonstrated that CityEngine is a potent tool for enhancing urban walkability, substantiated by a holistic approach that integrates a literature review, empirical data, and stakeholder feedback. The use of CityEngine for visualization and implementation has been instrumental in urban planning, aiding in decision-making and fostering stakeholder involvement. Notably, interventions informed by variable analysis, surveys, and interviews have led to wider sidewalks, new bike lanes, and pedestrian-centric features, thereby improving not just the infrastructure but also the perception of safety and the overall pedestrian experience. The creation of parks and the inclusion of public seating have enriched the public space, enabled greater recreational and social opportunities, and fostered a sense of community. Safety and accessibility have seen remarkable improvements, underscoring the importance of strategic urban design elements like crosswalks, signage, and lighting.

These enhancements align with sustainable urban development objectives, promoting eco-friendly transport options and supporting the evolution toward greener cities. The findings suggest a need for policies that back pedestrian-oriented designs and highlight the value of community involvement in urban planning. The ground-work laid by this study calls for future research to track the enduring impacts of these enhancements, assess their scalability, and examine the broader economic and social benefits of increased walkability. In sum, this research affirms that targeted design interventions, rooted in comprehensive environmental and community analyses, can significantly elevate walkability, and CityEngine's application in this realm highlights its potential to inform the development of safe, accessible, and sustainable urban environments. As we look ahead, it's imperative to maintain this progress, ensuring urban development is forward-thinking, inclusive, and responsive to the changing fabric of urban life.

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